

Moriah College

Mathematics Extension 1

2016 Preliminary Examination

General Instructions

- Reading time - 5 minutes
- Working time - 2 hour
- Write using black or blue pen
- Board-approved calculators may be used

Total marks - 70

Section I

10 marks

- Attempt Questions 1-10
- Allow about 20 minutes for this section

Section II

60 marks

- Attempt Questions 11-14

Section I

10 marks

Attempt Questions 1 - 10

Allow about 20 minutes for this section

Answer each question on the multiple choice answer sheet provided.

1)	If α and β are the roots of $2x^2 + 3x - 6 = 0$, what is the value of $\alpha\beta$? (A) $\frac{-3}{2}$ (B) 3 (C) $\frac{3}{2}$ (D) -3
2)	The quadratic equation $x^2 + kx + k = 0$ has its discriminant $\Delta = 0$. The value of k is: (A) 4 (B) 0 or 4 (C) 0 or -4 (D) 0
3)	A parabola has the equation $x^2 = 12(y - 2)$. The coordinates of the focus of this parabola are: (A) (0, 5) (B) (0, 3) (C) (0, 2) (D) (0, 1)
4)	Consider the function $y = \frac{4x^2 - 2x}{2x^2 + 2x}$ The value of $\lim_{x \rightarrow \infty} \left(\frac{4x^2 - 2x}{2x^2 + 2x} \right)$ is: (A) ∞ (B) 2 (C) 0 (D) 4

5)	Again, consider the function $y = \frac{4x^2 - 2x}{2x^2 + 2x}$ The value of $\lim_{x \rightarrow 0} \left(\frac{4x^2 - 2x}{2x^2 + 2x} \right)$ is: (A) 0 (B) ∞ (C) -1 (D) 1
6)	If $y = (x^3 + 1)^5$, then $\frac{dy}{dx} =$ (A) $15x^2(x^3 + 1)^4$ (B) $5(x^3 + 1)^4$ (C) $5(3x^2 + 1)^4$ (D) $15x^2(3x^2 + 1)^4$
7)	For a certain function, it is known that $\frac{dy}{dx} = (x - 1)^2(x - 2)$. The function has a minimum turning point at $x = 2$. At $x = 1$, the function has which of the following shapes: (A)  (B)  (C)  (D) 
8)	The function $y = 2x^3 - 24x + 1$ is concave down for (A) $x < 0$ (B) $x > 0$ (C) $-2 < x < 2$ (D) $x < -2$

9)	The quadratic equation $x^2 + 6x + k = 0$ has roots $x = \alpha$ and $x = 2\alpha$ What is the value of k? (A) $k = -16$ (B) $k = 8$ (C) $k = -2$ (D) $k = -8$
10)	It can be proven by mathematical induction that: $2^{n+2} + 3^{2n+1}$ is always divisible by a certain number x for $n \geq 1$ Which of the following numbers could be the value of x? (A) $x = 2$ (B) $x = 3$ (C) $x = 5$ (D) $x = 7$

Section I Questions 11-14 60 marks

Question 11 (15 marks)

- (a) Solve the equation $x^3 + \frac{8}{x^3} = 9$ using the substitution $M = x^3$ 3

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- (b) For what values of k is the quadratic expression $kx^2 + 4x + k$ positive for all values of x (ie positive definite). 3

-
- (c) Consider the function

$$y = \frac{2x + 1}{4x - 3}$$

Find the equation of the tangent to this curve at the point where $x = 1$ 3

-
- (d) α and β are the roots of the quadratic equation $x^2 - x - 1 = 0$

- (i) Find the value of $\alpha^2\beta + \beta^2\alpha$ 2

- (ii) Show that $\alpha^2 + \beta^2 = 3$ 2

- (iii) Find the value of $\alpha^3 + \beta^3$ 2

Question 12 (16 marks)

(a) Consider the function $y = x^4 - 18x^2$

(i) Find the x -coordinates of all stationary points of this function. 2

(ii) Find the x -coordinates of all points of inflexion of this function. 2

(iii) Sketch the curve of this function, showing the coordinates of the stationary and inflexion points. 3

(iv) Hence, find the values of k such that the function $y = x^4 - 18x^2 + k$ has no x -intercepts. 1

(b) Consider the function $y = x\sqrt{6-x}$

(i) Show that the derivative is given by: 2

$$\frac{dy}{dx} = \frac{12-3x}{2\sqrt{6-x}}$$

(ii) By finding any turning points, determine the maximum and minimum values of the function for the restricted domain $2 \leq x \leq 6$ 3

(c) A parabola has the equation $y = \frac{1}{8}x(x-8)$

(i) Find the vertex of this parabola. 2

(ii) Find the equation of the directrix of this parabola. 1

Question 13 (16 marks)

- (a) The point A is $(-2, 0)$ and the point B is $(6, 0)$.

The variable point P has coordinates (x, y)

- (i) Find an expression for the distance PB in terms of x and y . 1

- (ii) The point P moves so that the distance between P and B is three times the distance between P and A (ie $PB = 3PA$).

Show that the equation of the locus of P is given by the circle: 2

$$x^2 + 6x + y^2 = 0$$

- (iii) Find the centre of the circle. 1

(b)

- (i) Find the discriminant of the quadratic equation $x^2 - 2x + k = 0$. 1

- (ii) The line $y = 2x$ is a tangent to the parabola $y = x^2 + k$. 1
Draw a diagram showing the tangent and the parabola.

- (iii) Using part (i), find the value of k . 2

- (iv) Hence, find the coordinates of the point where the line touches the parabola. 2

-
- (c) Consider the following mathematical identity

$$\left(1 - \frac{1}{2^2}\right) \times \left(1 - \frac{1}{3^2}\right) \times \left(1 - \frac{1}{4^2}\right) \times \dots \times \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$$

- (i) Show that the identity holds true for $n=2$. 1

- (ii) Show that the identity holds true for $n=3$. 1

- (iii) Prove, by mathematical induction that the identity holds true for all positive integers $n \geq 2$. 3

- (iv) Using the identity with an appropriate value of n , find the value of 1

$$\frac{3}{4} \times \frac{8}{9} \times \frac{15}{16} \times \frac{24}{25} \times \dots \times \frac{9999}{10000}$$

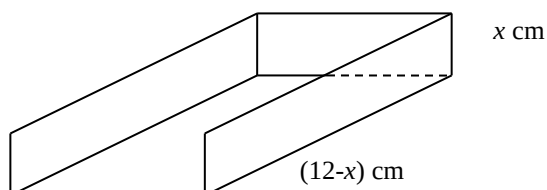
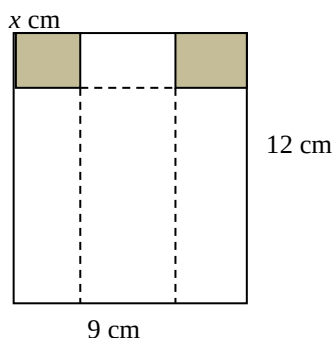
Question 14 (13 marks)

- (a) A rectangular sheet of cardboard measures 12cm by 9cm.

From two corners, squares of side x cm are removed as shown.

The remaining cardboard is folded along the dotted lines to form a tray as shown.

The height of the tray is x and the length of the tray is $(12-x)$.



- (i) Show that the volume, $V \text{ cm}^3$, of the tray is given by. 3

$$V = 2x^3 - 33x^2 + 108x$$

- (ii) Find the maximum volume of the tray 3

- (iii) Find the range of values that the height x can take, in order for the tray to be able to be constructed. 1

- (b) Let $P(x, y)$ be a variable point on the parabola $x^2 = 4ay$, where a is the focal length. Let $A(0, 4a)$ be a point on the y -axis.

- (i) Let D be the distance PA .

Show that $D^2 = y^2 - 4ay + 16a^2$ 3

- (ii) Show that the minimum value of D occurs when P is the point $(\pm 2\sqrt{2}a, 2a)$ 3

Ext 1
Solutions

MULTIPLE CHOICE

- | | | | |
|------|---|-------|---|
| (1.) | D | (2.) | B |
| (3.) | A | (4.) | B |
| (5.) | C | (6.) | A |
| (7.) | D | (8.) | A |
| (9.) | B | (10.) | D |
-

(11) (a) $M = x^3 \rightarrow M + \frac{8}{M} = 9$

$$M^2 - 9M + 8 = 0$$

$$(M - 8)(M - 1) = 0$$

$$M = 8 \text{ OR } 1$$

$$x^3 = 8 \text{ OR } 1$$

$$x = 2 \text{ OR } 1$$

(b) $\Delta = 16 - 4k^2$ and $a = k$

We need $16 - 4k^2 < 0$ and $k > 0$

$$k^2 > 4$$

$$k < -2 \text{ OR } k > 2 \text{ and } k > 0$$

$$\text{Hence } k > 2$$

$$(c) \quad u = 2x + 1 \quad ; \quad v = 4x - 3$$

$$u' = 2 \quad v' = 4$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{2(4x-3) - 4(2x+1)}{(4x-3)^2} \\ &= \frac{-10}{(4x-3)^2} \end{aligned}$$

$$\text{when } x = 1, \quad y = 3 \quad \text{and} \quad \frac{dy}{dx} = -10$$

$$\begin{aligned} \text{Tangent} \longrightarrow y - 3 &= -10(x - 1) \\ y &= -10x + 13 \end{aligned}$$

_____ u _____

$$\begin{aligned} (d) \quad (i) \quad \alpha\beta(\alpha + \beta) &= \frac{c}{a} \times \frac{-b}{a} \\ &= -1 \times 1 \\ &= -1 \end{aligned}$$

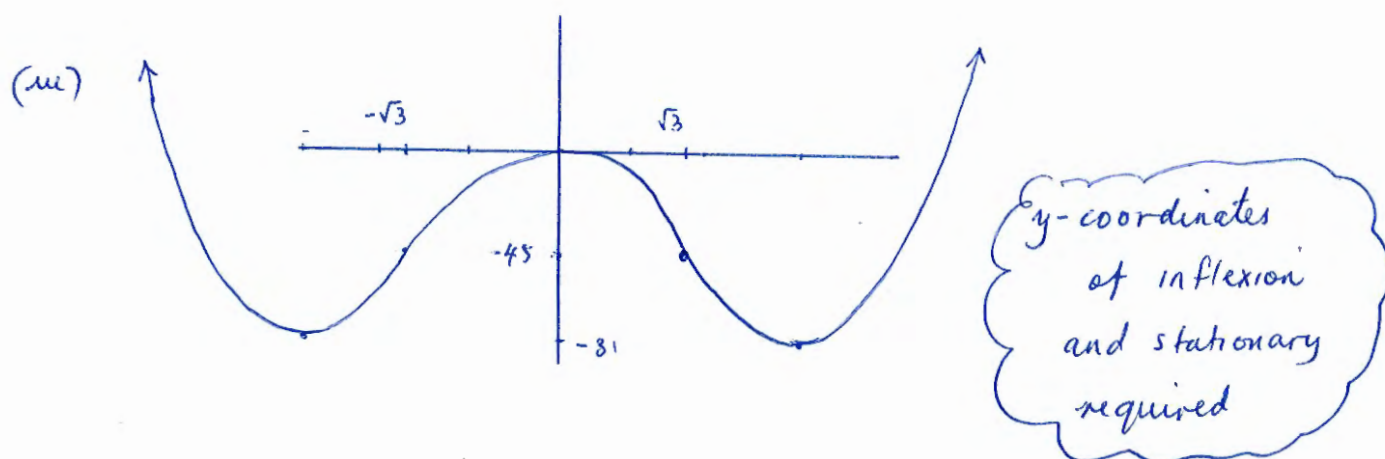
$$\begin{aligned} (ii) \quad \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= 1 - 2(-1) \\ &= 3 \end{aligned}$$

$$\begin{aligned} (iii) \quad (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2) &= 1(3 + 1) \\ &= 4 \end{aligned}$$

(12) (a) $\frac{dy}{dx} = 4x^3 - 36x$ and $\frac{d^2y}{dx^2} = 12x^2 - 36$

(i) $4x(x^2 - 9) = 0 \rightarrow$ stationary points when $x = 0, \pm 3$

(ii) $12(x^2 - 3) = 0 \rightarrow$ inflexion points when $x = \pm\sqrt{3}$



(iv) $k > 81$ to raise the curve above x-axis.

(b) (i) $y = x(6-x)^{1/2}$ $u = x$ $v = (6-x)^{1/2}$
 $u' = 1$ $v' = -\frac{1}{2}(6-x)^{-1/2}$
 $= \frac{-1}{2\sqrt{6-x}}$

$\frac{dy}{dx} = \sqrt{6-x} - \frac{x}{2\sqrt{6-x}} \rightarrow$ enough for full marks
 $= \frac{12 - 3x}{2\sqrt{6-x}}$

(u) $\frac{dy}{dx} = 0 \rightarrow x = 4$

$(4, 4\sqrt{2})$ is a max. turn. pt

$x = 2, y = 4$

$x = 6, y = 0$

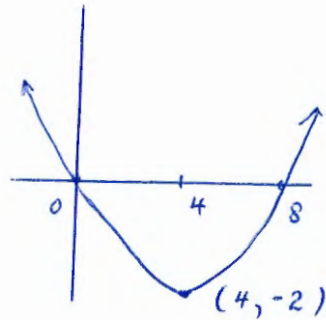
$\therefore$ Max. value is $4\sqrt{2}$

Min. value is 0

_____u_____

(c) $y = \frac{1}{8}x^2 - x$

(u) { vertex is $(4, -2)$ }



$4a = 8$

$a = 2$

(u) { Direction is $y = -4$ }

(13) (a)



(i) { distance PB = $\sqrt{(x-6)^2 + y^2}$ }

(ii) $\sqrt{(x-6)^2 + y^2} = 3\sqrt{(x+2)^2 + y^2}$

$$x^2 - 12x + 36 + y^2 = 9x^2 + 36x + 36 + 9y^2$$

$$8x^2 + 48x + 8y^2 = 0 \longrightarrow$$

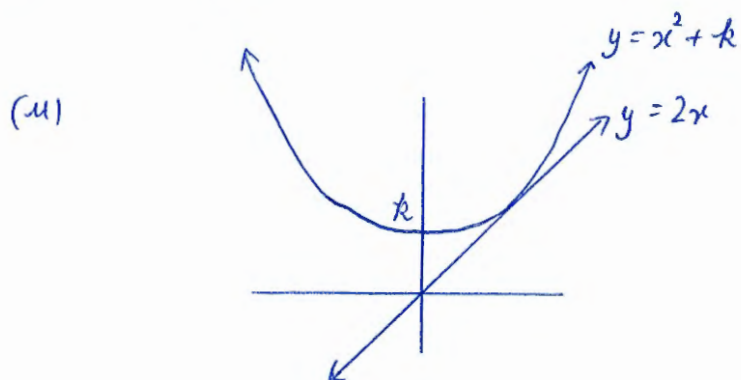
pay 2 marks to this point

{ $x^2 + 6x + y^2 = 0$ }

(iii) $(x+3)^2 + y^2 = 9$

{ centre is $(-3, 0)$ }

(b) (i) $\Delta = 4 - 4k$



(iii) $x^2 + k = 2x$ gives point(s) of intersection

$$x^2 - 2x + k = 0$$

$y = 2x$ is a tangent so $x^2 - 2x + k = 0$ has 1 root

$$\therefore 4 - 4k = 0 \longrightarrow \{ k = 1 \}$$

(iv) $x^2 - 2x + 1 = 0 \longrightarrow (x-1)^2 = 0 \longrightarrow \{ (1, 2) \}$

$$(c) (i) \quad n=2 \rightarrow LHS = 1 - \frac{1}{2^2} = \frac{3}{4}$$

$$RHS = \frac{2+1}{2(2)} = \frac{3}{4}$$

$$(ii) \quad n=3 \rightarrow LHS = \left(1 - \frac{1}{2^2}\right) \times \left(1 - \frac{1}{3^2}\right) = \frac{3}{4} \times \frac{8}{9}$$

$$= \frac{2}{3}$$

$$RHS = \frac{3+1}{2(3)} = \frac{2}{3}$$

$$(iii) \quad \text{Assume } \left(1 - \frac{1}{2^2}\right) \times \left(1 - \frac{1}{3^2}\right) \times \dots \times \left(1 - \frac{1}{k^2}\right) = \frac{k+1}{2k}$$

$$RTP \quad \left(1 - \frac{1}{2^2}\right) \times \dots \times \left(1 - \frac{1}{k^2}\right) \times \left(1 - \frac{1}{(k+1)^2}\right) = \frac{k+2}{2(k+1)}$$

$$LHS = \frac{k+1}{2k} \times \left(1 - \frac{1}{(k+1)^2}\right) \quad \text{by assumption}$$

$$= \frac{k+1}{2k} \times \frac{(k+1)^2 - 1}{(k+1)^2}$$

$$= \frac{k^2 + 2k}{2k(k+1)}$$

$$= \frac{k(k+2)}{2k(k+1)}$$

$$= \frac{k+2}{2(k+1)}$$

By principles of M.T.

true for $n \geq 2$

$n=2$ case proven in part (i)

$$(iv) \quad 1 - \frac{1}{n^2} = \frac{9999}{10000} \rightarrow n = 100$$

$\therefore$ Value of product is $\frac{101}{200}$

(14) (a)

(i) width of tray = $9 - 2x$

$$\begin{aligned}\text{Hence, } V &= x(12-x)(9-2x) \\ &= x(108 - 33x + 2x^2)\end{aligned}$$

$$V = 2x^3 - 33x^2 + 108x$$

(ii) $\frac{dV}{dx} = 6x^2 - 66x + 108$

solving $6x^2 - 66x + 108 = 0$

gives $x^2 - 11x + 18 = 0$

$$(x-9)(x-2) = 0$$

$$x = 2 \text{ or } 9 \quad (\text{only } x=2 \text{ is valid})$$

x	1	2	3
V'	+	0	-



→ $x = 2$ gives a max. volume

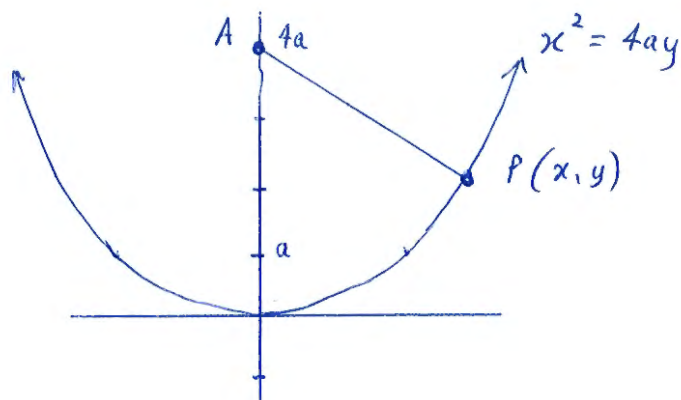
of $V = 2 \times 10 \times 5$

$$V = 100 \text{ cm}^3$$

(iii)

$$0 < x < \frac{9}{2}$$

(b) (c)



$$\begin{aligned} D^2 &= (x-0)^2 + (y-4a)^2 \\ &= x^2 + y^2 - 8ay + 16a^2 \end{aligned}$$

But $x^2 = 4ay$

$$\therefore D^2 = 4ay + y^2 - 8ay + 16a^2 \rightarrow \text{sufficient for full marks}$$
$$D^2 = y^2 - 4ay + 16a^2$$

(u) D^2 is a quadratic function in y (ie y is the variable)

$$\frac{d(D^2)}{dy} = 2y - 4a$$

solving $2y - 4a = 0 \rightarrow y = 2a$

Since D^2 is concave up, $\{y = 2a$ gives a minimum value of D^2 and hence D

Now when $y = 2a$, $x^2 = 4a(2a)$

$$x^2 = 8a^2$$

$$x = \pm 2\sqrt{2}a$$

$$\therefore P \text{ is } (\pm 2\sqrt{2}a, 2a)$$